COMSATS University Islamabad, Lahore Campus
Department of Computer Engineering

Course Information:

Course Title	Microprocessor Systems and Interfacing Lab
Course Code	CPE 342
Credit Hours	4(3,1)
Pre-requisite	CPE241 -DLD
Course In-charge	Engr. Moazzam Ali Sahi

Range of Complex Problem Solving (According to PEC Manual):

Range of Conflicting Requirement	Depth of Analysis Required	Depth of Knowledge Required	Familiarity of Issues	Extent of applicable codes	Extent of Stakeholder involvement	Consequences	Interdependence
☑	☑	☑	☐	☐	☐	☐	☐

Range of Complex Engineering Activities (According to PEC Manual):

Range of Resources	Level of interaction	Innovation	Consequences to society and the environment	Familiarity
☑	☑	☐	☑	☑

*Complex engineering problems must exhibit the characteristics of WP1 and some or all of WP2 to WP8.

Complex Engineering Problem Details:

Sr.#	Type	Attributes	Description
1.		Task	Design and implement an STM32-based embedded system for real-world applications including environmental monitoring, motor control, industrial automation, smart home control, data logging, or sensor-based safety systems. Students will integrate multiple peripherals using HAL libraries and develop a complete microcontroller solution demonstrating interfacing techniques, real-time processing, and human-machine interaction. **Design Requirements:** • Use STM32Fxxx microcontroller with HAL (Hardware Abstraction Layer) libraries • Interface at least **three different peripheral types** (e.g., sensors, actuators, displays, communication modules) • Implement at least **one communication protocol** (e.g., UART, SPI, I2C, PWM, ADC, or Timers)

			- Include appropriate signal conditioning and processing techniques for sensor data - Design a user interface using LCD/OLED display, LEDs, buttons, or serial communication - Implement interrupt-driven or timer-based real-time operation where applicable - Validate system performance through testing and measurement (response time, accuracy, reliability) - Document the complete hardware-software design process in a formal technical report - Present findings with live demonstration, circuit diagrams, code walkthrough, system testing results, and potential improvements	
2.	WP-1	Depth of knowledge required (Mandatory)	Requires in-depth knowledge of STM32 microcontroller architecture, HAL library functions, hardware interfacing protocols (UART, SPI, I_2C, PWM, ADC), interrupt handling mechanisms, timer configurations, GPIO operations, and real-time embedded system design principles. Students must understand datasheet specifications, signal conditioning circuits, and hardware-software integration beyond basic classroom instruction.	Breadth
3.	WP-2	Range of Conflicting requirements	Students must balance multiple conflicting technical requirements including timing constraints vs. processing overhead, power consumption vs. performance, interrupt-driven vs. polling-based approaches, hardware complexity vs. cost, real-time response vs. code maintainability, peripheral selection based on availability and compatibility, and memory limitations vs. functionality requirements. Design decisions require careful trade-off analysis.	
4.	WP-4	Depth of analysis required	Involves comprehensive analysis of timing diagrams, protocol selection justification (comparing UART vs. SPI vs. I_2C for specific applications), interrupt priority configuration, ADC sampling rate calculations, PWM frequency and duty cycle optimization, sensor interface circuit design, debouncing techniques evaluation, and system performance metrics (response time, accuracy, throughput). Requires abstract thinking to develop efficient solutions with no single obvious approach.	
10.	EA 1	Range of resources	Involves diverse resources including STM32Fxxx development boards, STM32CubeIDE or Keil MDK development environment, HAL library documentation, microcontroller datasheets, peripheral modules (sensors, actuators, displays), oscilloscopes and logic analyzers for signal verification, breadboards and testing equipment, online technical resources, application notes from ST Microelectronics, and research papers on embedded system design methodologies.	
11.	EA 2	Level of interaction	Requires team-based collaboration for task distribution (e.g., one member handles hardware interfacing, another focuses on firmware development, third on testing and	

			documentation), code integration and version control, troubleshooting hardware-software issues collectively, peer code reviews, and presenting technical demonstrations to instructors and peers. Involves interaction beyond internal team communication through technical presentations and documentation.
13	EA-4	Consequences to society and the environment	Students must evaluate the societal and environmental impact of their microcontroller-based system design. This includes analyzing power consumption and energy efficiency, assessing the use of electronic components and e-waste considerations, evaluating the system's contribution to automation and industrial productivity, examining safety implications in real-world deployments, considering accessibility and usability for diverse user groups, and justifying the project's significance in terms of societal benefits such as improved healthcare monitoring, environmental sensing, or industrial efficiency aligned with sustainable development goals and professional engineering ethics.
14	EA 5	Familiarity	The embedded system problems addressed may be unfamiliar to students, involving real-world industrial or commercial applications with specific timing requirements, integration of new peripheral modules not covered in standard labs, implementation of complex communication protocols, handling of asynchronous events and race conditions, and debugging of hardware-software interaction issues. Requires exploration, experimentation, and self-directed learning beyond classroom instruction.

United Nations Targeted SGDs (Sustainable Development Goals):

- **Goal 7 (Affordable and Clean Energy):** Microcontroller-based systems for energy monitoring, smart metering, and power management contribute to energy efficiency and optimization. STM32-based projects involving solar panel monitoring, battery management systems, or automated power control help ensure access to affordable, reliable, sustainable, and modern energy while reducing overall consumption through intelligent control systems.
- **Goal 9 (Industry, Innovation, and Infrastructure):** Embedded systems projects involving industrial automation, motor control, sensor networks, and IoT applications contribute to building resilient infrastructure and promoting inclusive and sustainable industrialization. STM32-based solutions for process automation, predictive maintenance, quality control, and smart manufacturing enhance industrial productivity and foster innovation in engineering sectors.
- **Goal 11 (Sustainable Cities and Communities):** Microprocessor-based systems for smart home automation, environmental monitoring (air quality, temperature, humidity), intelligent lighting control, waste management, and safety systems contribute to making

cities more inclusive, safe, resilient, and sustainable. Projects involving traffic monitoring, parking systems, or building automation support the development of sustainable urban infrastructure.

Assessment

CEP will mark against the terminal lab exam.

Performance Indicator	Marks	PLO*	Bloom's Taxonomy Domains/Levels		
			Cognitive	Psychomotor	Affective
1. Proposal Submission	5	PLO10			A2
2. Presentation	20	PLO10			A2
3. Demonstration (Hardware/Simulation)	30	PLO5		P3	
4. Viva	20	PLO3	C5		
5. Final Report	25	PLO10 PLO6 PLO7	C6		A2

Project Phases and Evaluation Criteria

Phase	Description	Weightage	Targeted PLO
1. Proposal Submission	Submit a problem statement, proposed STM32-based solution with block diagram, list of required peripherals and communication protocols, and expected outcomes.	5%	PLO-10
2. Presentation	Present project design includes hardware architecture, circuit diagrams, firmware flowcharts, peripheral interfacing methodology, and preliminary results/simulations.	20%	PLO-10
3. Demonstration	Show the actual working of the embedded system on STM32 hardware with live demonstration of all interfacing modules, real-time operation, and user interface functionality.	30%	PLO-5
4. Viva	Individual viva covering HAL library implementation, protocol selection justification, timing analysis, troubleshooting approaches, and design decisions.	20%	PLO-3
5. Final Report	Comprehensive technical documentation including problem definition, literature review, hardware design, firmware implementation, testing methodology, performance analysis, societal/environmental impact evaluation, and conclusions with future work recommendations.	25%	PLO-10